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Sensorless BLDC motor Sliding Mode Control for Disturbance Rejection on Hand Held Motor Drive

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Abstract

This research implemented a full sensor-less Sliding Mode Control (SMC) for a three phase Brushless Direct-Current (BLDC) motor for reduction on disturbance during motor operation. Instead of using hall sensors to provide motor speed feedback, this paper utilized the back counter-ElectroMotive Force (EMF), measured in the motor's control circuitry, and through the zero-crossing characteristic induced from its operation, a full feedback disturbance rejection is implemented for BLDC motor speed regulation. Experimental results compared this SMC control voltage demonstrated a 30% improvement on recovery time, as well as 25% reduction to the maximum disturbance, along with the traditional PI controllers. A full system identification of the motor parameters are also validated rigorously before selection of the SMC parameters. This result brings benefit of a low cost, firmware only efficient BLDC controller algorithm, that is automated in the Embedded system, for disturbance rejection and can be applied to a vast array of products that entails BLDC motors.

Keywords: Sensorless control, Sliding mode control, Embedded software, System identification

1 Introduction

BLDC motors replaced the commutating electric brush with metal-oxide-semiconductor field-effect transistor (MOSFET) electronics to rectify the DC current for smooth operation of the BLDC rotation. This design eliminated the physical contact of conducting components between the rotor and the stator, and in turn reduced motor friction, prevented electrical sparks and made space reduction available. Thus this mechanical advantage made BLDC motor more compact, quieter, and energy efficient compared to the brushed DC motor.

Closed loop control of motors requires a feedback signal, either with rotational sensors mechanically attached on the motor shaft, or with hall sensors installed onto the stator of the motor. Since mechanical sensors adds loading on the BLDC motor, and is hard to reduce its physical size, thus using a hall sensor to monitor the BLDC's rotation is much more preferred.

During recent development on utilizing motors, a thorough investigation had been reviewed regarding the use of motors in the Electrical Vehicle industry [1]. Extensive review in this research confirmed that torque ripple is the main concern as a disturbance for BLDC motors, and Sliding Mode Control (SMC) is one main aspect to deal with this disturbance [2]. On the other hand, BLDC are extensively used in household appliances as well, for example, one research focused on vacuum cleaner's BLDC motor, which face varying parameter change due to internal speed and temperature fluctuation over a wide range of motor speed operation[3].

Recently, many research focus on dealing with sensorless BLDC, reducing the reliance on Hall sensors for feedback. For example, a control observers are often considered to help estimate rotor state for BLDC motors [4], and it uses the back-EMF to provide the high order sliding mode observer to estimate the rotational speed of the motor. Alternatively, in another case, an adaptive saturation integral SMC controller was developed to deal with the speed fluctuation for small BLDC motors [5], and here a Kalman filter is used to design a torque observer for its SMC control.

Other controllers such as robust control are also designed to mitigate speed for BLDC motors, for example in [6], this research considers BLDC motors facing environments that bring variation to its parameter through load changes. There are also advanced two-step Linear Matrix Inequality (LMI) approach with SMC simulating BLDC motor control[7], trying to improve the chattering response common to SMC technique. Furthermore, there are attempts to deal with Permanent Magnet Synchronous Motor (PMSM) systems, also by using complicated reaching law SMC with an extended observer [8], where a feedforward disturbance observer is added to ensure stability, and also reduce chattering from SMC.

Many of the work mentioned above are based on past developments on Field Oriented Control (FOC) [9], where in BLDC and PSM motors the original three phase voltage and current is transformed into rotor phase angles, which adds computational load for the controller[10], and may require a more advanced controller Micro Processor Unit (MCU) for such a task.

Other control aspect of BLDC control also exists. To deal with motor ripple, one can manipulate its PWM voltage input waveform to conside both current and voltage for maximum torque production [11]. Some others consider a Fault Tolerant Control

(FTC) approach, to ensure motor current operation that keeps the temperature low for the motor operation[12]. While others consider a complex fuzzy PID controller strategy simulation by adding an artificial electrical field inside the motor[13]. Also some control simulation was tested on for BLDC using a frequency adaptive repetitive controller [14], where it automatically adapts PI controller parameter across different frequencies.

Apart from the control algorithm, there are hardware improvements to the general driver aspect to BLDC motors. Such as improvements on the motor drive high speed MOSFET transition in the BLDC switching [15], as well as applying artificial intelligence to re-design the PWM voltage input shape, to reduce torque ripples in BLDC motors [16].

As can be seen, different BLDC are applied in various environment which differs in motor size and power, where the control strategy is also influenced due to the computational power, as well as the main control concern. Here in this research the main goal is to develop an embedded control technology that is applied for a hand held motor drive device, such as a powered drill. The requirement here is that the algorithm must be fast and reliable, and the aim is to minimize the BLDC's speed disturbance as soon as possible. Therefore the embedded algorithm cannot be too complicated, such that it is operable for an already assigned Microprocessor Controller Unit (MCU). While the embedded scenario handles a sudden torque induced motor speed disturbance, as it is applied to the hand held device, and the recovery of speed feedback control is self regulated through the back EMF detection, providing future possibility to eliminate the hall sensor cost.

The paper is organized as follows. First, the motor model and SMC controller design is derived for both motor speed feedback and current monitoring in Sec. (2). Second, the BLDC motor parameters, its nominal value is validated through simulation, and its PID and SMC controller is designed accordingly in Sec. (3). Then experimentation for the motor drive is detailed for its results, followed by a short conclusion in the last part of the paper in Sec. (4). Followed with a short discussion in Sec. (5)

2 System Model and Controller Design

Depending on the BLDC phase construct, i.e. the number of electric magnets situated on the stator, respective MOSFET transistors are provided to switch accordingly in order to operate the BLDC motor. A three phase BLDC electric circuit is illustrated in Fig. (1) showing our implementation experiment study in this research.

Circuit input from this MOSFET arrangement utilize a six-step square wave that switches on and off the different MOSFETs, namely $A+$, $A-$, $B+$, $B-$, $C+$ and $C-$, to rotate the motor rotor in a cyclic order. Where the plus denotes input of current and the minus denotes outlet of current. Thus in this sequence the electrical magnets A , B and C are activated accordingly, to create a rotational magnetic field.

The six-step square wave I/O sequence is depicted in Fig. (2). Notice the Pulse Wave Modulated (PWM) bursts on the plus branches, as this modulates the mean voltage amplitude controlling the rotational speed.

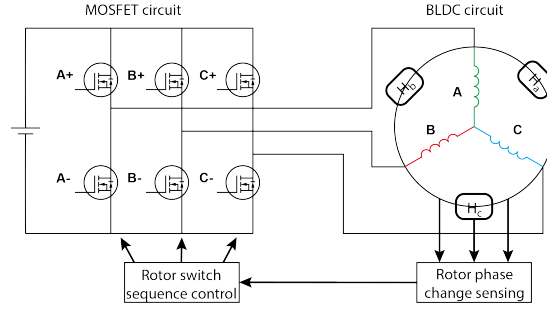


Fig. 1 Electrical circuit for a three-phase BLDC motor. Six MOSFET transistors are configured to switch the DC current inside the BLDC circuit consecutively on the branch A, B and C. In order to create a smooth rotation of the rotor.

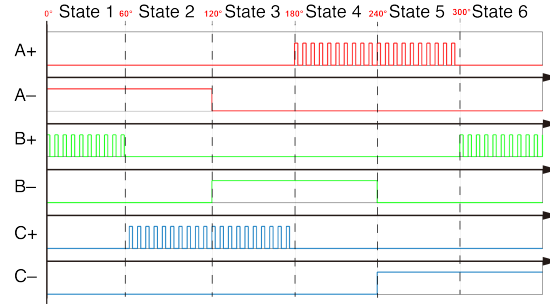


Fig. 2 Six step square pulse wave modulation voltage control switching time sequence for BLDC rotor .

However, BLDC motors have back EMF with trapezoid shapes during its three phase operation, as illustrated in Fig. (3).

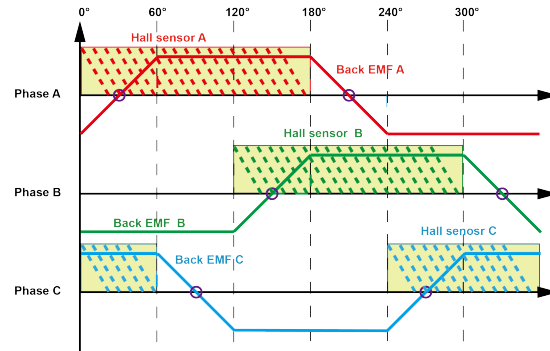


Fig. 3 Three positional signals provided by hall sensors, and its associated back EMF voltage measurement.

Back Electro-Motive Force (EMF) is the induced voltage in the motor circuitry when the rotor is commuting alongside the stator, arising from the conductance being

excited from the changing magnetic field as the rotor spins. Therefore this back EMF is observed in the motor circuitry, one can use it as a feedback signal.

The BLDC motor control system in an industrial setting is setup as shown in Fig. (4).

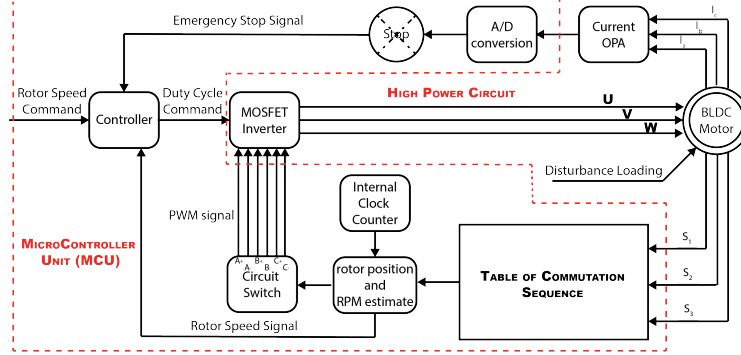


Fig. 4 BLDC motor control system schematics. Speed control reference is provided to the controller, which gives the inverter a duty cycle command as it in turn regulates the BLCD motor drive through the three phase circuit.

The three phase BLDC motor is shown at the upper right, where three input voltages U , V and W are sent to drive the motor. These three voltages are PWM sent from the inverter MOSFET where the correct circuitry of $A + / -$, $B + / -$ and $C + / -$ are chosen. The duty cycle of the PWM are also regulated from the controller input, such that the nominal voltage amplitude are determined for U , V and W .

With the three phase voltage input, the BLDC will induce rotation to the rotor, and hence its circuit can also be monitored for its three phase current I_a , I_b and I_c , that is illustrated on the upper side of the BLDC motor. Three currents are sent into an Operational Amplifier (OPA) for safety inspection, and then converted into digital form to trigger an emergency stop.

Another set of signals the BLDC provided are the rotor rotational signals. These can either be hall sensor signals that points out what angular position the rotor is currently located, or the sensor-less back EMF signal that provides the same information. In this control flow model, the two signals are considered the same output function.

Note that this rotational signal set are shown at the lower right corner of Fig. (4), and is sent into the table of commutation sequence. This table is illustrated for enlargement here in Tab. (1).

This commutating sequence is essential in driving the BLDC motor rotation smoothly. In our industrial MCU setting, only when the rotor has reached the desired position, as it is being identified from the table, will trigger the next commuting phase signal to orient the magnetic field accordingly. Thus sequentially “turn” the rotor to the next position. Note that the MCU decimal sequence does not exactly match the STATE, but has its own reference number from 1 ~ 6, as they are passed to the MCU for rotor speed estimation.

Table 1 Rotational signal (commutation) sequence conditioning

	Switching States						Position Signals			MCU Decimal
STATE	A+	A-	B+	B-	C+	C-	S_1	S_2	S_3	
State 1	0	1	1	0	0	0	0	1	1	3
State 2	0	1	0	0	1	0	0	1	0	2
State 3	0	0	0	1	1	0	1	1	0	6
State 4	1	0	0	1	0	0	1	0	0	4
State 5	1	0	0	0	0	1	1	0	1	5
State 6	0	0	1	0	0	1	0	0	1	1

At a high motor speed operation, for example a 4000 *rpm* operation, the rotor rotates one full cycle in 0.015 *sec*. Meaning the rotational speed is 418.88 *rad/sec*, and each phase switch occurs merely a 0.005 *sec*.

The challenge here is to both accurately identify the phase switching for the transistor commutation, and also correctly estimate the motor speed. This is achieved by utilizing the rotor positional signal that is provided from the BLDC motor. Recall that this signal can either be the hall sensor signal h_1 , h_2 and h_3 , or the back EMF signal b_1 , b_2 and b_3 . Therefore in Fig. (4) this signal is not distinguished but combined into S_1 , S_2 and S_3 positional signals.

Both commutation switching and rotor speed estimation requires precision timing in the system model. These are achieved by incorporating an internal Micro-controller (MCU) digital clock, that times exactly the three positional sensor signal transitions. The clock first provide the commutation command at the right time, the embedded system tallies the full rotational time for one cycle for RPM estimation. As shown in Fig. (4).

Finally the rotational position signal is fed back into the PWM Circuit Switch control to assign the correct circuit branch for MOSFET transistors to function, and the estimated rotor speed is used to compare with the reference signal. And in this research the SMC controller is tuned such that a disturbance is rejected for this hand-held device. Note that in Fig. (4) the MCU unit consists of components within the red dashed line, and are implemented by software (digitally), and the other components belongs to the high power circuit board or the BLDC motor itself.

2.1 BLDC motor/electric system

The mathematical model for BLDC controller design relies on the combined equations of its voltage, mechanical, back EMF and torque equation.

For our study of a three phase BLDC, the two phase rotor ($p = 2$) is rotating inside the stator, and its voltage governing equation shows that the current i in the circuit is coupled between the three electric magnets, performing as inductances L in the circuitry, as well as the resistance R . And by driving the input voltages (U , V , W) into the motor (Fig. (1)), the voltage equation is shown as followed.

$$\mathbf{R} \cdot \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \mathbf{L} \cdot \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} = \begin{bmatrix} U \\ V \\ W \end{bmatrix} \quad (1)$$

And in our case of a symmetrical three phase BLDC motor, the resistance and inductance matrix are defined as $\mathbf{R} = \text{diag}\{R, R, R\}$, $\mathbf{L} = \text{diag}\{L, L, L\}$, where $L = L_{\text{self}} - L_{\text{mutual}}$ considers only the difference between self and mutual inductance due to symmetry.

$[e_a \ e_b \ e_c]^T$ in Eq.(1) is the back EMF term that reflects the induced generated voltage arising as the rotor moves against the 120° separated inductances. This term is determined according to the magnetic flux projection factor value ($f_a(\theta_e)$, $f_b(\theta_e)$ and $f_c(\theta_e)$), and considers the EMF's magnitude as the product of the rotor's rotational speed ω_e , the magnetic influx density λ_m as it is projected according to its angular function.

$$\begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} = \lambda_m \omega_e \begin{bmatrix} f_a(\theta_e) \\ f_b(\theta_e) \\ f_c(\theta_e) \end{bmatrix} \quad (2)$$

A table is created for the functional projection f_a , f_b and f_c in Tab. (2). This table outlines the three phase EMF projection response for our BLDC motor. Where the roator angle in associated to the different states, as well as its projected flux.

Table 2 Magnetic flux projection value due to rotor mechanical rotational angle.

No.	θ_e	$f_a(\theta_e)$	$f_b(\theta_e)$	$f_c(\theta_e)$
1	$0^\circ \sim 60^\circ$	1	-1	$1 - \frac{6}{\pi}\theta_e$
2	$60^\circ \sim 120^\circ$	1	$\frac{6}{\pi}\theta_e - 3$	-1
3	$120^\circ \sim 180^\circ$	$\frac{6}{\pi}\theta_e + 5$	1	-1
4	$180^\circ \sim 240^\circ$	-1	1	$\frac{6}{\pi}\theta_e - 7$
5	$240^\circ \sim 300^\circ$	-1	$9 - \frac{6}{\pi}\theta_e$	1
6	$300^\circ \sim 360^\circ$	$\frac{6}{\pi}\theta_e - 11$	-1	1

Note that for each state, one of the flux projection in phase a , b or c is linearly proportional to the electric angle as in $6/\pi\theta_e$, as the projection first grow, then stays steady and finally declines when the rotor moves in and out of range relative to a phase inductance. Recall that this trapezoid shape functions are illustrated earlier in Fig. (3).

Provided the electro-magnetic torque is defined as total power in circuit to drive the mechanical rotational speed ω_m , in our three phase case it is shown as:

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m}.$$

Since the rotor electrical angle θ_e is half the pole number of the mechanical angle θ_m , it relates the electrical and mechanical speed as $\dot{\theta}_e = \omega_e = \frac{P}{2}\omega_m$, it can be shown

that the electro-magnetic torque is derived as:

$$T_e = \frac{P}{2} \lambda_m [i_a f_a(\theta_e) + i_b f_b(\theta_e) + i_c f_c(\theta_e)]. \quad (3)$$

Where P is the pole number, and in our case $P = 2$. i_a , i_b and i_c are the phase current associated (shown in Fig. (1) and Fig. (4)).

Thus this is finally related for the mechanical drive in the three phase BLDC motor rotational equation:

$$T_e = J_m \cdot \frac{d}{dt} \omega_m + B_m \cdot \omega_m + T_L. \quad (4)$$

Where J_m is the moment of inertia of the motor, B_m is the damping ratio (friction) and T_L is the loading torque.

2.2 Feedback Controller design

The BLDC motor is controlled through firmware programming in this system, with its loop closed by feeding back the rotational speed.

Two controllers, a classical PI control and a Sliding Mode Control (SMC) controller is implemented for comparison. The derivation of the sliding mode is outlined here in this section, in comparison for the PI performance. With uncertainty, the SMC controller drives the system state error e and its differential $\dot{e}$ towards zero, in order to achieve successful reference input tracking.

The model uncertainty given are chosen in the inertia parameter J and its damping ratio B , that implicitly takes the loading variation T_L disturbance into account for a straight forward controller derivation. This is illustrated in following equation:

$$T_e = (J + \Delta J) \cdot \frac{d}{dt} \omega_m + (B + \Delta B) \cdot \omega_m + T_L. \quad (5)$$

Where ΔJ and ΔB are the uncertain moment of inertia and damping ratio. Thus in order to drive the output convergence on input reference, a rotor speed command system with its sliding surface defined as

$$S = e(t) = \omega^* - \omega_m, \quad (6)$$

where ω^* is the command and ω_m is its actual speed. It is clear that the surface is guaranteed to drive Lyapunov function V to zero for reference convergence, as it is defined:

$$V = \frac{1}{2} \cdot S^2, \quad \dot{V} < 0 \quad for \quad S \neq 0.$$

The command U is then derived from the dot product of Lyapunov V , shown here:

$$\dot{V} = S \dot{S} = S \left[\frac{d}{dt} \omega^* - \frac{1}{J_m} (-B_m \omega_m - T_L + T_e) \right] < 0. \quad (7)$$

Thus by utilizing the electrical torque T_e to overcome the unmodeled T_L , we choose to make T_e as the command u in the maximum switching form as:

$$T_e = u = u_0 \text{sign}(S), \quad (8)$$

where it is clear that the sign function is used to drive u accordingly

$$\text{sign}(S) = \begin{cases} 1, & \text{if } S > 0 \\ -1, & \text{if } S < 0 \end{cases}$$

With this disturbance rejecting command, one can derive the minimum u_0 from substituting Eq. (8) into Eq. (7).

$$\begin{aligned} \dot{V} &= S \left[\frac{d}{dt} \omega^* + \frac{B_m}{J_m} \omega_m + \frac{T_L}{J_m} \right] - \frac{u_0}{J_m} S \cdot \text{sign}(S) \\ &= S \left[\frac{d}{dt} \omega^* + \frac{B_m}{J_m} \omega_m + \frac{T_L}{J_m} \right] - \frac{u_0}{J_m} \cdot |S| < 0 \end{aligned}$$

It is clear that one would choose the value of u_0 to be

$$u_0 > \left| J_m \frac{d}{dt} \omega^* + B_m \omega_m + T_L \right|. \quad (10)$$

provided when J_m and B_m both includes some positive uncertainty, with the following range:

$$\begin{cases} 0 < a < (J_m = J + \Delta J) < b \\ 0 < c < (B_m = B + \Delta B) < d \end{cases} \quad (11)$$

Thus it is clear the selection of u_0 is to guarantee the following specification:

$$\begin{cases} u_0 > \left| a \cdot \frac{d}{dt} \omega^* + c \cdot \omega_m + T_L \right|, \\ u_0 > \left| a \cdot \frac{d}{dt} \omega^* + d \cdot \omega_m + T_L \right|, \\ u_0 > \left| b \cdot \frac{d}{dt} \omega^* + c \cdot \omega_m + T_L \right|, \\ u_0 > \left| b \cdot \frac{d}{dt} \omega^* + d \cdot \omega_m + T_L \right|. \end{cases} \quad (12)$$

Since it is desired to keep the reference speed ω^* constant $\omega^* = \text{const.}, \forall t > 0$ at all times, which meant $\frac{d}{dt} \omega^* = 0$, and this simplifies Eq. (12) into:

$$\begin{cases} u_0 > |c \cdot \omega_m + T_L|, \\ u_0 > |d \cdot \omega_m + T_L|. \end{cases} \quad (13)$$

And since $d > c$, finally the SMC command parameter will have u_0 chosen as:

$$u_0 > |d \cdot \omega_m + T_L|. \quad (14)$$

The final control input Eq. (14) is a concise calculation that considers minimal parameters, this implies that there is minimum calculation needed, and therefore the most important parameter to design is the value of d , where it follows Eq. (11).

3 Motor identification for embedded system

The SMC controller is implemented for a commercially available Milwaukee Tool MP18-CPD handheld drill. The internal electronics control board was originally provided by Princeton Technology, and in order to develop the SMC controller, this research used the Princeton Technology's MCU-CK2532 development board, that has all the onboard mosfet power electronics, as well as a MCU processing unit. In order to tune for optimal control parameter, the feedback controller design, including both traditional PI control, as well as the proposed SMC control, all simulation was done in MATLAB, and later compared with experiment test.

3.1 Motor system hardware and electronics

The BLDC motor under control is inside a hacked commercial hand held drill from Milwaukee Tool M18-CPD, which is annotated in Fig. (5). The whole system setup that realize the SMC controller is illustrate in Fig. (6) and the MCU board photo is shown in Fig. (7).

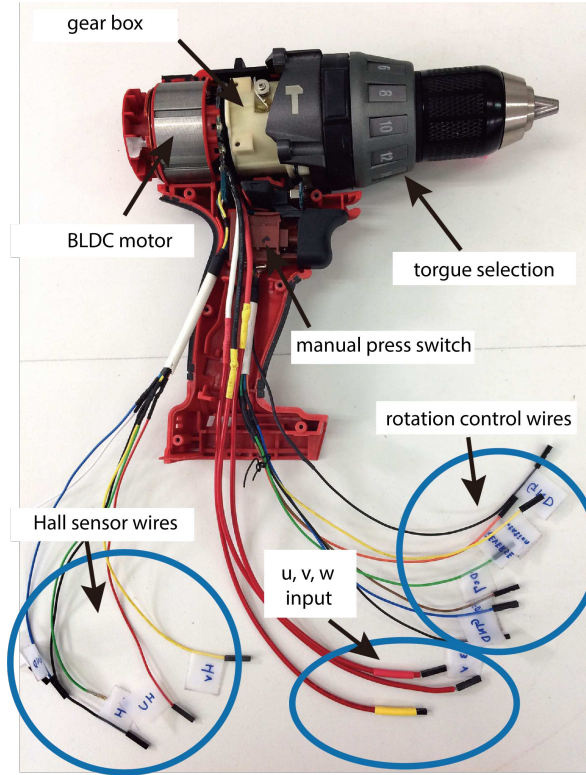


Fig. 5 Hacked hand held Milwaukee Tool M18-CPD power drill with labeled wire extensions.

As shown here in Fig. (5), the electronics wiring is extended to connect to the MCU-CK2532 board in Fig. (7), where both hall sensor wires, as well as voltage input wires are seen in the lower-left and lower corners. Where the three phase (A, B, C) hall sensor signals, and the motor control u, v, w voltage input are used respectively. On the lower right corner are miscellaneous wires related to the rotational direction switch, start/stop wirings for the M18-DPD. Note that the back-EMF signal is derived from taking the difference of input control reference voltage from the actual monitored voltage, and is instantaneously found on the MCU-CK2532 circuitry.

The full controller system setup that drives the BLDC control is shown here in Fig. (6), showing the components on board the MCU-CK2532. These includes the MCU (PT32M622), Princeton Technology's 32 bit ARM based microcontroller, a gate drive (FAN7382) from Fairchild, as well as the MOSFETs (PT4N60) from Princeton Technology. Finally the whole board controls the POWERSTATE™ BLDC motor from Milwaukee® Tool.

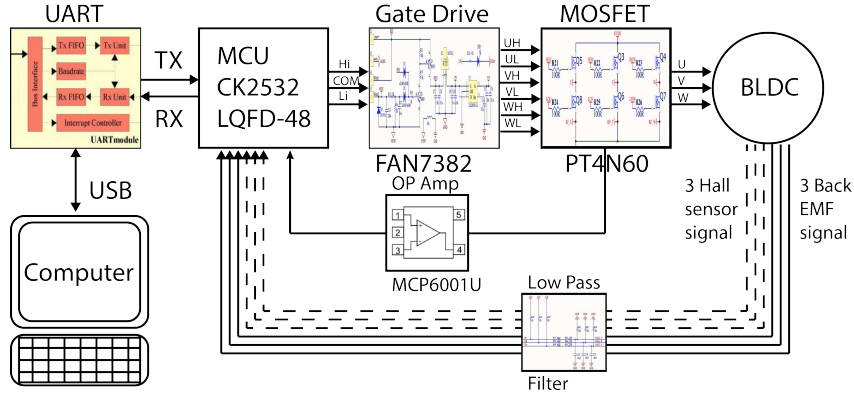


Fig. 6 System setup of the BLDC control for hacked Milwaukee Tool MP18-CPD with the MCU-CK2532 board, showing various part of the board, including MCU, Gate Drive, and other circuit involved.

As can be seen, a personal computer is used to code and upload the control algorithm through UART into the MCU, such that during real-time control, the MCU directly pass the command reference signal voltages, as well as taking the feedback signal provided from the circuit low pass filter. While the gate drive commands the six MOSFET to deliver the correct BLDC circuit drive, the MCP6001U is used to monitor the current flow in the MOSFET circuitry. Thus in total the MCU takes feedback signals for all three phase voltages (u, v, w), current (i_u, i_v, i_w), as well as hall sensor voltage (h_a, h_b, h_c).

The connections of the above mentioned 3 phase voltage, current, and hall sensor signal can be seen here on the MCU board (MCU-CK2532) here in Fig. (7), outlined with the different components, as indicated. There are also some other additional connections, for example on top of the board, a HVDC connection is the supply input of the high-voltage direct current used for the MOSFETs. Next to HVDC is the GND connection used for the board ground.

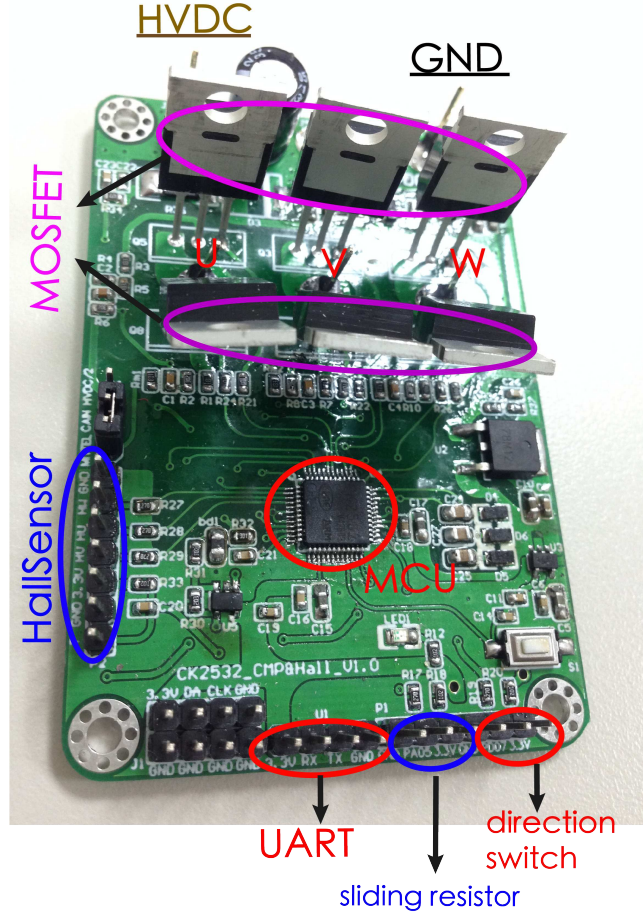


Fig. 7 Princeton Technology's MCU-CK2532 development Embedded system board.

And on the bottom of the board are the UART connections, as well as a sliding resistor and direction switch, as manual inputs when the hand held devise is in use. However, these are override during our experiments.

3.2 Motor and controller parameter used for embedded system

As expected, the speed regulating feedback control, for both traditional PI control and SMC control, needs to be designed according to the BLDC motor parameters, as mentioned in Eq. (11). Such that the coefficients a, b, c , and d are chosen correctly.

The following table outlines the Milwaukee®POWERSTATE™ BLDC motor's design parameter. As one can see, most of the important parameters such as $J_m = J + \Delta J$ and $B_m = B + \Delta B$, are already provided from the manufacturer specifications. Thus providing grounds for choosing parameter a, b, c and d . Where the most important value of $d = 0.0015$ is used in our case.

Table 3 Motor parameter used for system ID and embedded system

Name	Symbol	Value	Units
Terminal Resistance	R_s	0.12	Ω
Terminal Inductance	L_s	80	μH
Rotor Inertia	J_m	0.000203 ± 0.0005	$kg \cdot m^2$
Rotor Damping	B_m	0.001 ± 0.0005	$N \cdot m \cdot s/rad$
Nominal Voltage	HVDC	18	V
Max. Torque	T_e	70	$N \cdot cm$
Back EMF const.	λ_m	8.8×10^{-3}	$V \cdot s/rad$
Battery Capacity		4.0	Ah
Rotor Poles	P	4	

Source: Milwaukee.

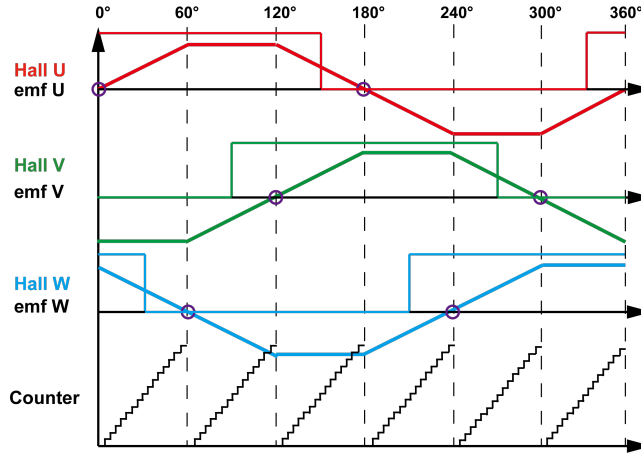
The original commercial controller that is in use are PI controllers, and its equation is well known as:

$$u[n] = K_p e[n] + K_i \sum_{i=1}^n e[i]. \quad (15)$$

Where in consideration, one has to tune both K_p and K_i according to its performance required.

3.3 Back-EMF motor speed estimation

In order to accurately control and reject motor speed disturbances from T_L variation, which is introduced from the hand drill usage, ω_m needs to be estimated. In this research, an instantaneous counter is used to keep track of the time passed between zero-crossings of the Back-EMF occurrences, or the hall sensor switches. As illustrated in the following Fig. (8). It can be seen in Fig. (8) that a counter k accumulates at

**Fig. 8** Normalized counter usage for motor speed calculation.

start of a back-EMF crossing detection, for example the U back-EMF detection signal,

as shown in read on the upper axis, and ends its count when encountering the next zero-crossing incident, e. g. the blue W EMF located at the third axis. The counts are then registered to $T_i[k]$, and k is reset to zero to count further until a full rotation cycle is finished. By summing the different T_i counts from $i = 1 \sim 6$ calculates the time spent rotating a full cycle. ω_m is then calculated using the following equation:

$$\omega_m = \frac{60}{3} \frac{f}{\sum_{i=1}^6 T_i P}. \quad (16)$$

Where the constant 60 is used to transform RPS to RPM, and the constant $1/3$ denotes when alternating zero-crossing occurrences happen at 120° of the full 360° stator. Again, P is the pole number of the rotor, denoting the P number of zero-crossing changes for one full rotor cycle. Note that in this research experiment, a fixed sample rate of $f = 200$ kHz is used.

4 Results

Both simulation and experiment are carried out for comparison of the performance of the proposed SMC controller, especially against the traditional PI controllers. The following table outlines the controller parameters used for both simulation and experiment. The PI controller parameters k_P and k_I are manually tuned, while the SMC coefficient u_0 is selected based on Eq. (14).

Table 4 Simulation and Experimental controller parameters

Controller	Simulation	Experiment
PI1	$k_p = 0.06, k_i = 0.01$	$k_p = 0.0125, k_i = 0.0078$
PI2	$k_p = 0.1, k_i = 0.08$	$k_p = 0.025, k_i = 0.007$
SMC	$u_0 = 17$	$u_0 = 17.4$

Note that the controller coefficient is different between the simulation and experimental, as there are some un-modeled variance that did not show up on simulation. However, the main difference between PI1 and PI2 controller is that PI1 is tuned with no overshoot, but PI2 allows for overshoot to exist.

The result of PI1 controller with feedback is shown here in Fig. (9), comparing the simulation results and its experimental result, responding to a step input of 4000 RPM command.

It can be seen that the experimental (actual red crosses) corresponds well to the simulation (blue solid line), showing traditional PI controllers holding steady at 4000 RPM, reaching steady at 1.5 seconds (rise time). PI2 controller response is shown next in Fig. (10), with the same step input of 4000 RPM.

As can be seen, both experiment and simulation respond similarly with a slight overshoot before reaching its steady state of 4000 RPM, albeit the experimental data having some delay compared to the simulation results. Also fig. (9) and fig. (10)

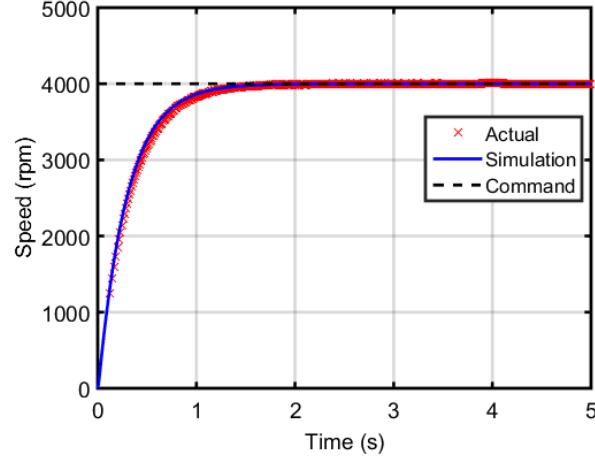


Fig. 9 Comparison of PI1 controller performance responding to a step input of 4000 RPM, for both simulation and experiment using EMF sensorless feedback.

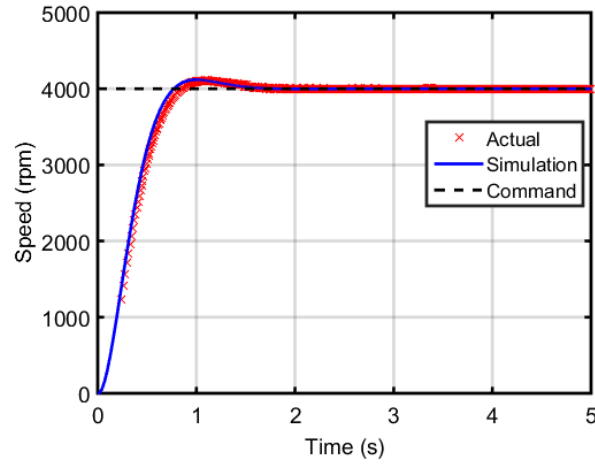


Fig. 10 Comparison of PI2 controller performance responding to a step input of 4000 RPM, for both simulation and experiment using EMF sensorless feedback.

demonstrated that simulation and experiment verified the system identification in Tab. (3), providing confidence in using the SMC controller.

Fig. (11) shows the response of simulation and experiment of the proposed SMC controller, also driving a step input of 4000 RPM. Similar to the previous two figures, the result shows an identical trend between the experimental and simulation results, verifying the control algorithm and system identification.

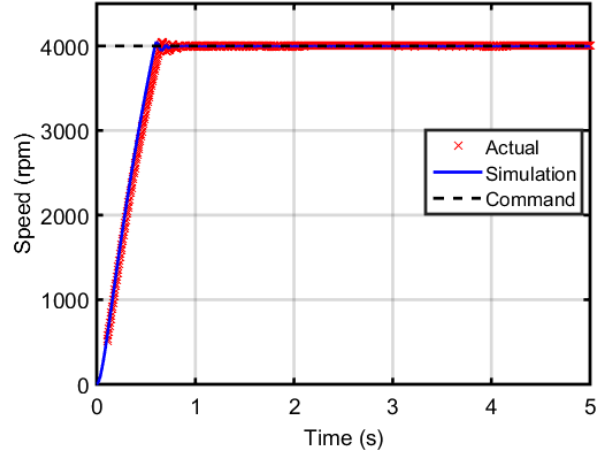


Fig. 11 Comparison of SMC controller performance responding to a step input of 4000 RPM, for both simulation and experiment using EMF sensorless feedback.

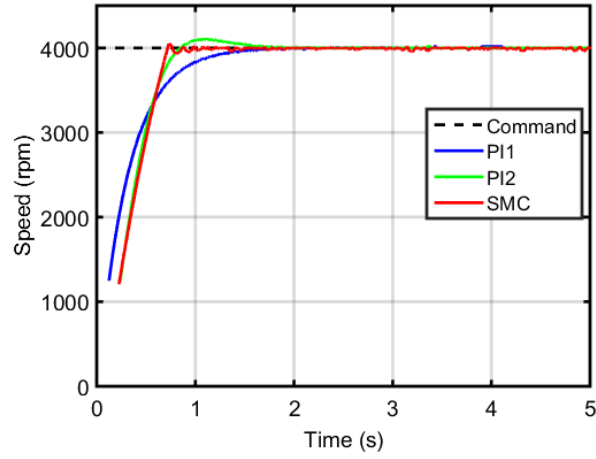


Fig. 12 Comparison of experimental results on different controller performance for step input using EMF sensorless feedback.

Fig. (12) compares the performance of the previous three controllers, and shows that the SMC controller has the least rise time compared to the PI1 and PI2 controllers, as well as a smaller overshoot before achieving its steady state response.

Our SMC controller is applied here towards an disturbance rejection scenario, where a fixed torque disturbance is introduced through the drill. The torque disturbance is reflected as a ω_m drop in motor speed, where the SMC controller is required to compensate as soon as possible.

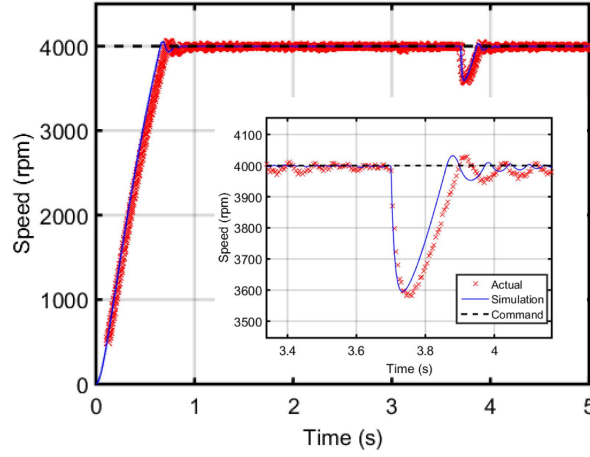


Fig. 13 Speed variation comparison for disturbance rejection using SMC controller between simulation and experiment.

Fig. (13) shows the full extent of the experiment, where the hand held device is first driven into a steady state of 4000 RPM, then around 3.65 sec, the drill is forced into a wooden block at a fixed depth, so the motor speed is dropped to around 3600 RPM, then the controller quickly compensates it back to the set-point of 4000 RPM.

Fig. (13) also has an enlarged proportion of the disturbance rejection, showing ω_m response between 3.4 ~ 4.9 second. Both experimental and simulation response are depicted here. As can be seen, the disturbance is an (inverse) impulse input, and thus the controller compensates it with first some over-compensation, then later some residual oscillation but ultimately returns to a steady state.

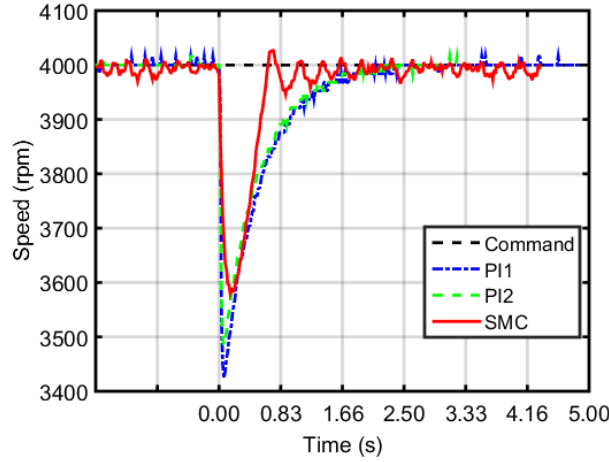


Fig. 14 Comparison on different controller rejection result due to disturbance using EMF sensorless feedback.

Finally, in Fig. (14), the different controllers' disturbance rejection performance is shown side by side. As can be seen, the disturbance is rejected best using the SMC controller as expected, where both PI1 and PI2 both having a larger maximum drop in ω_m variation, and takes longer time to recover to the original set-point. The following Tab. (5) shows the performance improvement of SMC controller comparing the settling time, overshoot percentage for the step input response, as well as the disturbance rejection's recovery time and maximum disturbance.

Table 5 Motor disturbance rejection time response characteristics comparison

controller	settling time (s)	overshoot (%)	recovery time (s)	max disturbed(%)
PI1	1.653	N/A	0.913	14.25
PI2	0.812	3.11	0.851	12.75
SMC	0.682	1.25	0.581	10.25

Recovery time is defined as time measured for its impulse response duration.

It is clear that SMC controller outperform both PI controllers. In the case for step input, both PI1 and PI2 have longer 5% settling time, while SMC is less than half the settling time for PI1 controller. Remember that PI1 is tuned so that there are no overshoot, while PI2 has a faster response experiencing a 3% overshoot. But SMC still outperform PI2 with half its overshoot.

In the case of disturbance rejection, this research defines the recovery time as the time when ω_m deviates from 2.5% set-point, i.e. the time between leaving and returning to 3900rpm. It can be seen that SMC has the fastest recovery time, about half of the PI1 performance. SMC also outperforms all PI controllers with minimized disturbance, maxing at about 10% deviation.

5 Discussion

This research presented a low cost SMC feedback controller design replacing the commercial PI controllers in embedded systems, that outperforms the torque disturbance rejection often required in hand held devices. Using both simulation and experiment, our SMC controller outperforms the worst case PI both in recovery time and maximum deviation. The promising use of this simple SMC shows that is no need to replace the current MCU embedded unit to an higher cost device, and only requires a software upgrade, and can be incorporated into the next product cycle easily.

In addition to the improvement on SMC controller in the embedded system, another contribution in this paper is the use of the Back-EMF ω_m estimation. While this voltage monitoring add some additional cost but can be offset by eliminating of the hall sensors installation, or act as an additional verification for speed estimation, as it shows validity through the paper's investigation.

Future direction for continuing this work includes adding an software torque observer for this BLDC model, while it is hard to actually implement a real torque sensing setup, but can be estimated through theoretical calculation. Another extension of this work is to apply this SMC controller to a Permanent Magnetic Synchronised

Motor (PMSM), where it should be a straight forward task, since only the back EMF shape is changed from a trapiziod to a sinusoid. Also, the use of Field Oriented Control (FOC) in BLDC and/or PMSM motor through a advanced MCU device is worth investigating on the effect of combining SMC with FOC. Last but not least, both BLDC and PMSM motors with current i control added to this SMC control is also worth some investigation.

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